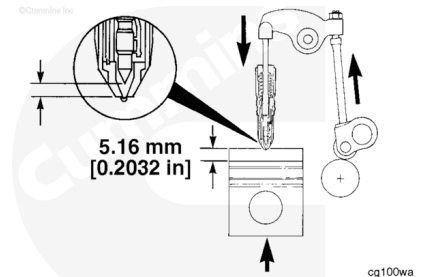


Trainings ▾

General Information

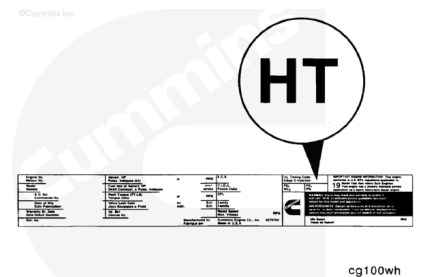
The injection timing is the relative measurement of the distance remaining between the injector plunger and the injector cup when the piston is 5.16 mm [0.2032 in] or 19 degrees before top dead center on the compression stroke.

Note : Injector timing is expressed by the amount of push rod travel remaining.



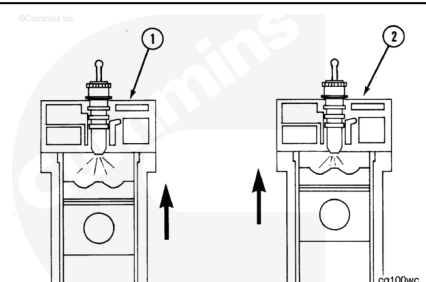
The injection timing code appears on the engine dataplate. Codes are alphabetic letters that relate to a numerical specification.

Specifications can be found in the Control Part List (CPL) Manual, Bulletin 4021328.



Note : The next six frames are a brief review of injection timing and how it can be adjusted.

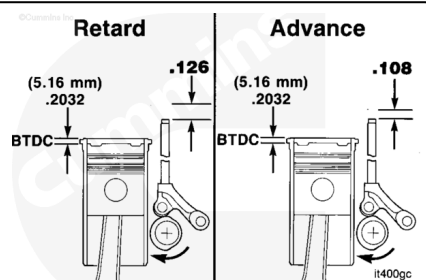
Advanced timing (1) means the fuel is injected earlier into the cylinder during the compression stroke. Retarded timing (2) means the fuel injection occurs closer to top dead center in the cylinder.



The amount of pushrod travel determines the time of fuel injection in relation to the piston position.

A low numerical value of the pushrod travel remaining indicates a greater degree of advanced (1) or fast timing.

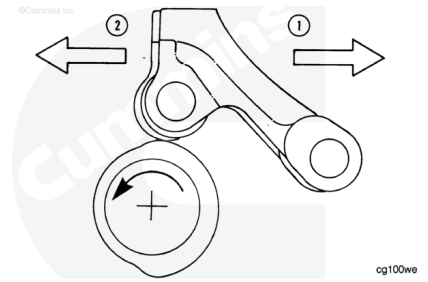
A high numerical value of pushrod travel remaining indicates a greater degree of retarded (2) or slow timing.



Injection timing changes are accomplished by advancing or retarding the cam follower action in relation to the piston position.

This is accomplished by changing the orientation of the camshaft lobe to the cam follower using different camshaft gear keys.

Note : Gear train timing (index mark alignment) always remains the same.

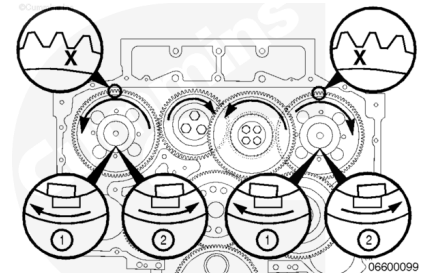


Note : The direction of normal rotation on a QSK78 engine crankshaft is clockwise as viewed from the front.

The camshaft key provides a means of indexing the camshaft with the gear.

Offset keys allow the camshaft profile to be rotated slightly while the gear train timing remains the same.

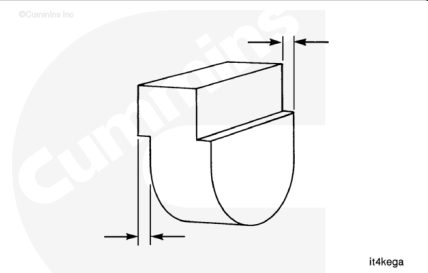
The more the top of the offset is moved in the direction of the camshaft normal rotation, the more the injection timing will be retarded. The pushrod travel numerical value will increase.



Note : If checking or setting the injection timing, it is recommended to use a testing gear. A testing gear is a camshaft gear that has been modified to provide a slip-fit on the camshaft.

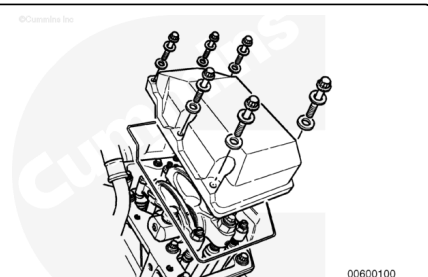
Offset keys can be identified by measuring the offset and referring to the chart at the end of this section.

Note : Each 0.025 mm [0.001 in] of offset will cause a 0.0127-mm [0.0005-in] change in the pushrod travel from a straight key.

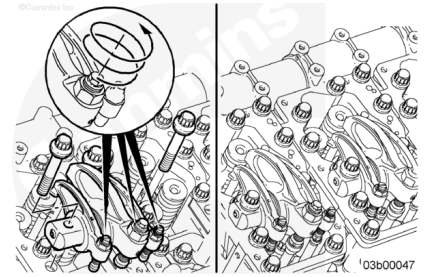


Preparatory Steps

Remove the rocker lever cover from cylinder number 1 on the right bank. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/50/50-003-011.html>).

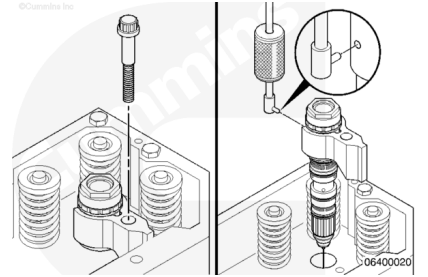


Remove the rocker lever assembly from cylinder number 1 on the right bank. Refer to Procedure 003-009 in Section 3 (</qs3/pubsys2/xml/en/procedures/50/50-003-009.html>).



Remove the injector from cylinder number 1 on the right bank. Refer to Procedure 006-026 in Section 6 (</qs3/pubsys2/xml/en/procedures/50/50-006-026-tr.html>).

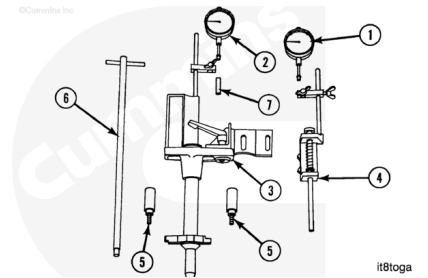
Note : It is **not** necessary to remove all injectors; however, engine rotation will be easier if all of the injectors are removed.



Setup

Use the injection timing tool, Part Number 3824942, and QSK78 adapter kit, Part No. 3164219. The indicators (1 and 2) are identical.

1. Push rod travel indicator
2. Piston travel indicator
3. Piston plunger support assembly
4. Push rod plunger support assembly
5. Hold-down adapter
6. Extension assembly (adapter wrench)
7. Indicator stem extension.



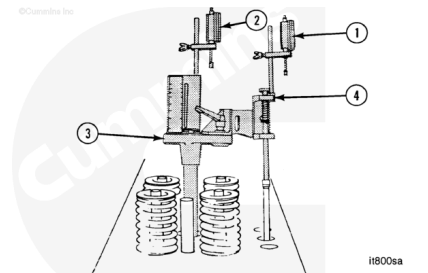
The pushrod plunger support assembly alignment is critical.

Install the pushrod plunger support (4) in the outside slot of the piston plunger support (3).

Align the pushrod plunger support with the mark. Tighten the capscrew.

Install the indicators (1 and 2) on the posts. Turn the indicators so they are **not** over the plungers.

Install the stem extension on the piston travel indicator.

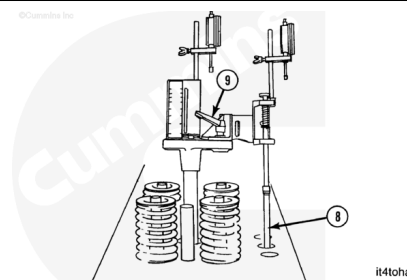


Install the injector pushrod (8) for cylinder number 1 on the right bank.

Install the timing tool in the injector bore of cylinder number 1 on the right bank. Install the hold-down adapters.

Make sure the pushrod plunger and the pushrod are aligned.

Tighten the support lock (9).



Measure

⚠ CAUTION ⚠

Only use the crankshaft to rotate the engine. If any other gear is used, a false measurement will result. Gear lash must be closed up in the direction of normal rotation.

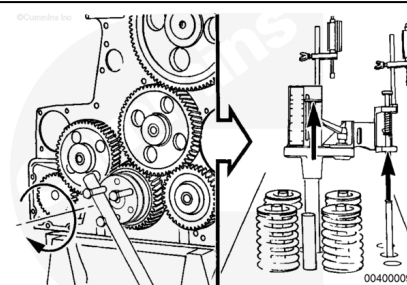
Note : For detailed information on the timing marks. Refer to Procedure 001-008 in Section 1 (</qs3/pubsys2/xml/en/procedures/50/50-001-008.html>).

Note : Three guide bolts equally spaced in front of the crankshaft will help rotate the engine.

Turn the crankshaft in the direction of normal rotation while observing both of the timing tool plungers. Both plungers will begin moving upward when the cylinder is on the compression stroke.

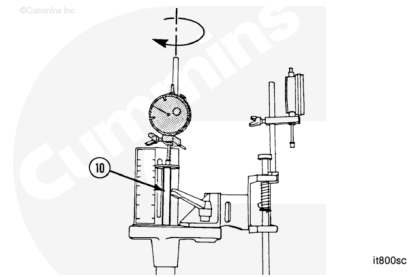
Note : Assuming all the gear index marks were aligned when the injection timing process was started, the crankshaft will have to be rotated approximately $\frac{3}{4}$ of a revolution to get to the compression stroke for cylinder number 1 on the right bank.

If both plungers are **not** moving up (one up and one down), the engine is on the exhaust stroke. Rotate the crankshaft one revolution to get on the compression stroke.



Establish top dead center by slowly rotating the crankshaft in the direction of normal rotation while observing the piston plunger (10). The plunger will move upward, stop, then begin to move downward. The stop point of the plunger is top dead

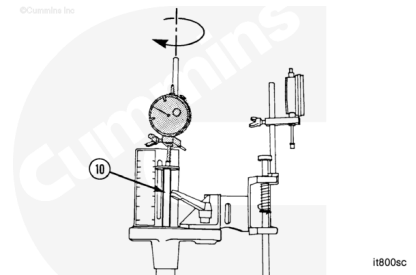
center. Rotate the engine opposite the direction of normal rotation until the plunger begins to move down. The cylinder is now slightly before top dead center.



Turn the piston travel dial indicator so that the stem is touching the plunger. Carefully move the indicator downward until the needle has turned a minimum of five revolutions 12.7 mm [0.500 in]. Lock the piston dial indicator in position.

Slowly turn the crankshaft in the direction of normal rotation until the piston dial indicator needle stops turning **clockwise**, top dead center (TDC).

Once top dead center has been established the piston dial indicator position needs to be reset to make sure there is sufficient range to measure the piston position at 19 degrees before TDC. Move the piston dial indicator downward until there is **only** one revolution, 2.54 mm [0.100 in], of travel remaining until the indicator bottoms out. Lock the piston dial indicator in position.

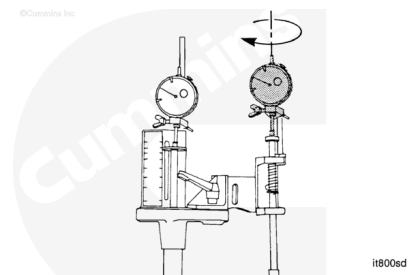


Adjust the indicator to zero.

Turn the pushrod dial indicator so that the stem touches the plunger.

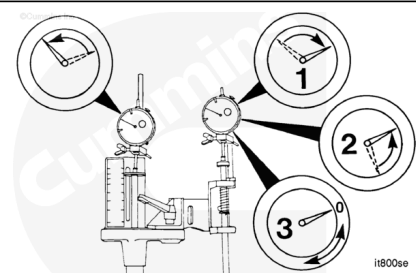
Carefully lower the indicator until it bottoms out. Raise the dial indicator until the needle has turned a minimum of three revolutions, 7.62 mm [0.300 in].

Lock the push-rod dial indicator in position.



Slowly turn the crankshaft in the direction of normal rotation until the pushrod indicator stops (1), momentarily reverses direction (2) (this is the crush nose on the camshaft), and stops again (3). The cam follower is now on the outer base circle of the camshaft. The piston is now approximately 45 degrees after TDC.

Once this position has been established the push rod dial indicator position needs to be reset to make sure there is sufficient range to measure the push rod remaining to measure the push rod remaining travel at 19 degrees before TDC.



⚠ CAUTION ⚠

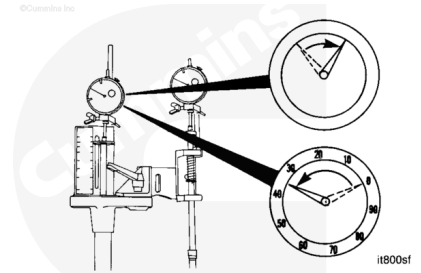
It is important to record the amount of travel remaining in the pushrod travel indicator for later reference.

Carefully lower the push rod travel indicator until it bottoms out. Raise the indicator approximately $\frac{1}{2}$ of a revolution 1.27 mm [0.050 in]. Lock the indicator in position.

Set the indicator at zero.

Rotate the engine in the opposite direction of normal rotation while observing the piston dial indicator.

Stop rotating the crankshaft when the piston travel indicator indicates the piston is at top dead center (zero). Check TDC (zero) again.



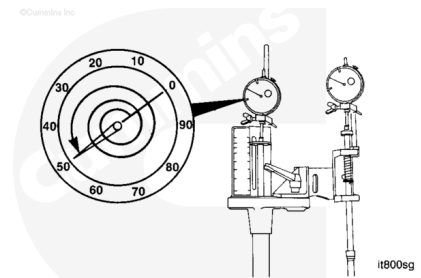
⚠ CAUTION ⚠

The crankshaft must be turned slowly to count the indicator revolutions accurately.

Turn the crankshaft opposite the direction of normal rotation until the indicator piston travel dial needle moves $2\frac{1}{2}$ revolutions or 6.35 mm [0.250 in].

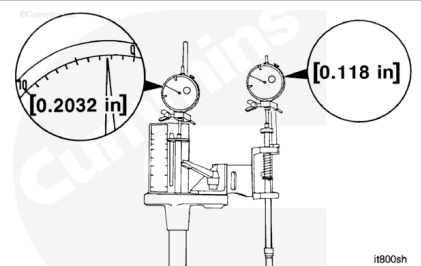
The piston is now 6.35 mm [0.250 in] before TDC.

Rotate the crankshaft in the direction of normal rotation until the piston ravel dial indictor needle reads 5.16 [0.203 in]. The piston is now in the correct position to measure the remaining push rod travel



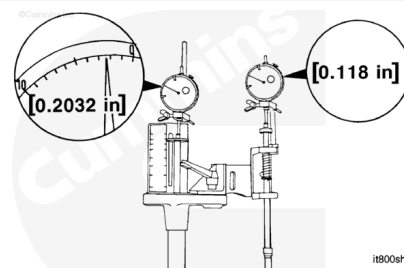
⚠ CAUTION ⚠

Only move the piston to 5.161 mm[0.203 in] before top dead center by turning the crankshaft in the direction of normal rotation. If the crankshaft is turned too far, the crankshaft must be turned in the opposit ein the direction of normal rotation more than 5.161 mm [0.203 in] before TDC. Very slowly turning the crankshaft in the direction of normal rotation until the indicator indicates that the piston is 5.161 mm [0.203 in] before TDC.



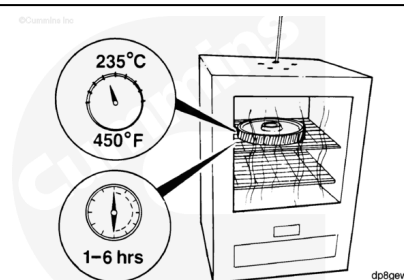
Note : Remember that the QSK78 injection timing specifications are more than one indicator revolution, 2.54 mm [0.100 in].

Read the pushrod travel indicator **counterclockwise** from zero. This is the injection timing measurement to compare to the specification. An example of 3 mm [0.118 in] is shown.



If **not** sure of the number of pushrod indicator revolutions, check by carefully lifting the indicator stem until the indicator has bottomed out. Lower the stem the amount of excess travel that was set in the third preceding step. Lower the stem to the plunger. Read the indicator.

If the injection timing is within specification and a slip-fit gear is used, install the standard gear. Refer to Procedure 001-012 in Section 1 (</qs3/pubsys2/xml/en/procedures/50/50-001-012.html>). Repeat the injection timing procedure after the camshaft gear has cooled.



⚠ CAUTION ⚠

To avoid engine damage, the timing measurement must be confirmed after changing the key.

If the injection timing is **not** within specification, repeat the measurement procedure to check the tool setup and the zero settings.

If the timing is still **not** within specification, the camshaft key **must** be changed. For instructions to remove the camshaft gear. Refer to Procedure 001-012 in Section 1. (</qs3/pubsys2/xml/en/procedures/50/50-001-012.html>)

Record the orientation of any offset of the key. Use the following worksheet to determine an alternate key.

Note : Repeat this procedure on a front and rear cylinder of the bank(s) from which the camshaft or camshaft key was replaced.

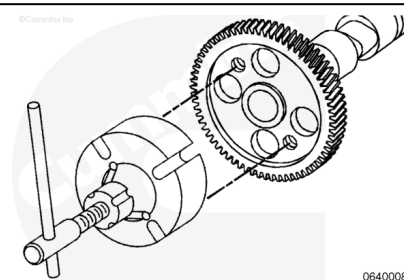


Table C																
S-302	200704	200706	200708	200709	200711	200714	216294	216782	3000491	3000492	3000493	3000494	3000495	Part Number	Direction of Offset	
0.0256	0.0351	0.0411	0.0366	0.0326	0.0289	0.0441	0.0273	0.0311	0.0391	0.0436	0.0471	0.0491	0.0512	3000495	Opposite	
0.0235	0.033	0.039	0.0345	0.0305	0.0268	0.042	0.0252	0.029	0.037	0.0415	0.045	0.0471	0.0491	3000494	Opposite	
0.0215	0.031	0.037	0.0325	0.0285	0.0248	0.04	0.0232	0.027	0.035	0.0395	0.043	0.0441	0.0471	3000493	Opposite	
0.0185	0.028	0.034	0.0295	0.0255	0.0218	0.037	0.0202	0.024	0.032	0.0365	0.04	0.042	0.0441	200714	Opposite	
0.018	0.0275	0.0335	0.029	0.025	0.0213	0.0365	0.0197	0.0235	0.0315	0.036	0.0395	0.0415	0.0436	3000492	Opposite	
0.0155	0.025	0.031	0.0265	0.0225	0.0188	0.034	0.0172	0.021	0.029	0.0335	0.037	0.039	0.0411	200706	Opposite	
0.0135	0.023	0.029	0.0245	0.0205	0.0168	0.032	0.0152	0.019	0.027	0.0315	0.035	0.037	0.0391	3000491	Opposite	
0.011	0.0205	0.0265	0.022	0.018	0.0143	0.0295	0.0127	0.0165	0.0245	0.029	0.0325	0.0345	0.036	200708	Opposite	

Table C																
0.0095	0.019	0.025	0.0205	0.0165	0.0128	0.028	0.0112	0.015	0.023	0.0275	0.031	0.033	0.0351	200704	Opposite	
0.007	0.0165	0.0225	0.018	0.014	0.0103	0.0255	0.0087	0.0125	0.0205	0.025	0.0285	0.0305	0.0326	200709	Opposite	
0.0055	0.015	0.021	0.0165	0.0125	0.009	0.024	0.0072	0.011	0.019	0.0235	0.027	0.029	0.0311	216782	Opposite	
0.0033	0.0128	0.0188	0.0143	0.0103	0.006	0.0218	0.005	0.009	0.0168	0.0213	0.0248	0.0268	0.0289	200711	Opposite	
0.0017	0.0112	0.0172	0.0127	0.0087	0.0051	0.0202	0.0034	0.0072	0.0152	0.0197	0.0232	0.0252	0.0273	216294	Opposite	
0	0.0095	0.0155	0.011	0.007	0.003	0.0185	0.0017	0.006	0.0135	0.018	0.0205	0.0235	0.0256	S-302	N/A	
0.0017	0.0078	0.0138	0.0093	0.0053	0.0016	0.0168	0	0.0038	0.0118	0.0163	0.0298	0.0218	0.0239	216294	Same	
0.0033	0.0062	0.0122	0.0077	0.0037	0	0.0152	0.0016	0.002	0.0102	0.0147	0.0282	0.0202	0.0223	200711	Same	
0.0055	0.004	0.01	0.0055	0.0015	0.002	0.013	0.0038	0	0.008	0.0125	0.026	0.018	0.0201	216782	Same	
0.007	0.0025	0.0085	0.004	0	0.0037	0.0115	0.0053	0.0015	0.0065	0.011	0.0245	0.0165	0.0186	200709	Same	
0.0095	0	0.006	0.0015	0.0025	0.0062	0.009	0.0078	0.004	0.004	0.0085	0.02	0.014	0.0161	200704	Same	
0.011	0.0015	0.0045	0	0.004	0.0077	0.0075	0.0093	0.0055	0.0025	0.007	0.0105	0.0125	0.0146	200708	Same	

Table C

0. 01 35	0. 00 4	0. 00 2	0. 00 25	0. 00 65	0. 01 02	0. 00 5	0. 01 18	0. 00 8	0	0. 00 45	0. 00 8	0. 01	0. 01 21	30 00 49 1	Sam e
0. 01 55	0. 00 6	0	0. 00 45	0. 00 85	0. 01 2	0. 00 3	0. 01 38	0. 01	0. 00 2	0. 00 25	0. 00 6	0. 00 8	0. 01 01	20 07 06	Sam e
0. 01 8	0. 00 85	0. 00 25	0. 00 7	0. 01 1	0. 01 47	0. 00 05	0. 01 63	0. 01 25	0. 00 45	0	0. 00 35	0. 00 55	0. 00 76	30 00 49 2	Sam e
0. 01 85	0. 00 9	0. 00 3	0. 00 75	0. 01 15	0. 01 52	0	0. 01 68	0. 01 3	0. 00 5	0. 00 05	0. 00 3	0. 00 5	0. 00 71	20 07 14	Sam e
0. 02 15	0. 01 2	0. 00 6	0. 10 5	0. 01 45	0. 01 82	0. 00 3	0. 01 98	0. 01 6	0. 00 8	0. 00 35	0	0. 00 2	0. 00 41	30 00 49 3	Sam e
0. 02 35	0. 01 4	0. 00 8	0. 01 25	0. 01 65	0. 02 02	0. 00 5	0. 02 18	0. 01 8	0. 01	0. 00 55	0. 00 2	0	0. 00 21	30 00 49 4	Sam e
0. 02 56	0. 01 61	0. 01	0. 01 46	0. 01 86	0. 02 23	0. 00 71	0. 02 39	0. 00 2	0. 01 21	0. 00 76	0. 00 41	0. 00 21	0	30 00 49 5	Sam e

Answer each of the following questions in the spaces provided. The answers to the questions and the use of Tables A, B, and C will determine the timing key required to correct the injection timing.

A working example is attached for your review to illustrate the use of this worksheet.

1.	What is the current timing?	_____
2.	What is the timing code?	_____
3.	What is the timing specification for this code ($\pm$ 0.002 in)?	_____

4.	Is the current timing a larger or smaller number than the specification? • If larger, advance the timing. • If smaller, retard the timing.	
5.	What is the difference between the current timing (answer to question 1) and the specification (answer to question 3)?	
6.	Does the offset of the current key point in the same or opposite direction that the camshaft normally rotates?	
7.	Use Table A to determine the current key part number. • What is the amount of the offset of the current key? • What is the part number of the current key?	

Table A

Timing Key Part Number	Offset	
	mm	in
200704	0.500	0.0197
200706	0.833	0.0328
200708	0.584	0.0230
20709	0.381	0.0150
200711	0.178	0.0070
200714	0.991	0.0390
216294	0.089	0.0035
216782	0.279	0.0110
3000491	0.686	0.0270
3000492	0.914	0.0360
3000493	1.092	0.0430
3000494	1.194	0.0470
3000495	1.295	0.0510

Table A			
Timing Key Part Number	Offset		
	mm	in	
3626781	0.229	0.0090	
3626782	0.305	0.0120	
3626783	0.432	0.0170	
3626784	0.546	0.0215	
8.	Use Table B to determine how to use Table C. Circle or check the appropriate answer.		
Table B			
Answer to Question 4	Answer to Question 6	Beginning Point on Table C	
Advance	Same	Top of column	_____
Advance	Opposite	Bottom of column	_____
Retard	Same	Bottom of column	_____
Retard	Opposite	Top of column	_____
9.	Answer the following questions before using Table C to determine the new timing key part number. - Find the current key part number listed at the top of the column on Table C. - Move up or down the column (the answer to question 8). Do not pass 0.000 (zero).		

If you pass 0.000 (zero), you will be choosing a key that does the opposite of what you want it to do.

Stop when you locate the number nearest (± 0.002 in) to the required change in push rod travel (answer to question 5). Remain in this row. Move you finger to the right. The result is the new key part number and direction of offset the timing key **must** point.

- What is the part number of the new key?

Note : Each column on Table C indicates the change in the push rod travel. The change will result if the key at the top of the column is removed and the new key indicated in the second column from the right is installed. The direction of offset will be pointing as indicated in the last column to the right on Table C.

Table C

S-302	200704	200706	200708	200709	200711	200714	216294	216782	3000491	3000492	3000493	3000494	3000495	Part Number	Direction of Offset
0.0256	0.0351	0.0411	0.0366	0.0326	0.0289	0.0441	0.0273	0.0311	0.0391	0.0436	0.0471	0.0491	0.0512	3000495	Opposite
0.0235	0.033	0.039	0.0345	0.0305	0.0268	0.042	0.0252	0.029	0.037	0.0415	0.045	0.0471	0.0491	3000494	Opposite
0.0215	0.031	0.037	0.0325	0.0285	0.0248	0.04	0.0232	0.027	0.035	0.0395	0.043	0.0441	0.0471	3000493	Opposite
0.0185	0.028	0.034	0.0295	0.0255	0.0218	0.037	0.0202	0.024	0.032	0.0365	0.04	0.042	0.0441	200714	Opposite
0.018	0.0275	0.0335	0.029	0.025	0.0213	0.0365	0.0197	0.0235	0.0315	0.036	0.0395	0.0415	0.0436	3000492	Opposite
0.0155	0.025	0.031	0.0265	0.0225	0.0188	0.034	0.0172	0.021	0.029	0.0335	0.037	0.039	0.0411	200706	Opposite
0.0135	0.023	0.029	0.0245	0.0205	0.0168	0.032	0.0152	0.019	0.027	0.0315	0.035	0.037	0.0391	3000491	Opposite
0.011	0.0205	0.0265	0.022	0.018	0.0143	0.0295	0.0127	0.0165	0.0245	0.029	0.0325	0.0345	0.036	200708	Opposite

Table C

0.0095	0.019	0.025	0.0205	0.0165	0.0128	0.028	0.0112	0.015	0.023	0.0275	0.031	0.033	0.0351	200704	O p p o s i t e
0.007	0.0165	0.0225	0.018	0.014	0.0103	0.0255	0.0087	0.0125	0.0205	0.025	0.0285	0.0305	0.0326	200709	O p p o s i t e
0.0055	0.015	0.021	0.0165	0.0125	0.009	0.024	0.0072	0.011	0.019	0.0235	0.027	0.029	0.0311	216782	O p p o s i t e
0.0033	0.0128	0.0188	0.0143	0.0103	0.006	0.0218	0.005	0.009	0.0168	0.0213	0.0248	0.0268	0.0289	200711	O p p o s i t e
0.0017	0.0112	0.0172	0.0127	0.0087	0.0051	0.0202	0.0034	0.0072	0.0152	0.0197	0.0232	0.0252	0.0273	216294	O p p o s i t e
0	0.0095	0.0155	0.011	0.007	0.003	0.0185	0.0017	0.006	0.0135	0.018	0.0205	0.0235	0.0256	S-302	N/A
0.0017	0.0078	0.0138	0.0093	0.0053	0.0016	0.0168	0	0.0038	0.0118	0.0163	0.0298	0.0218	0.0239	216294	Sa m e
0.0033	0.0062	0.0122	0.0077	0.0037	0	0.0152	0.0016	0.002	0.0102	0.0147	0.0282	0.0202	0.0223	200711	Sa m e
0.0055	0.004	0.01	0.0055	0.0015	0.002	0.013	0.0038	0	0.008	0.0125	0.026	0.018	0.0201	216782	Sa m e
0.007	0.0025	0.0085	0.004	0	0.0037	0.0115	0.0053	0.0015	0.0065	0.011	0.0245	0.0165	0.0186	200709	Sa m e
0.0095	0	0.006	0.0015	0.0025	0.0062	0.009	0.0078	0.004	0.004	0.0085	0.02	0.014	0.0161	200704	Sa m e
0.011	0.0015	0.0045	0	0.004	0.0077	0.0075	0.0093	0.0055	0.0025	0.007	0.0105	0.0125	0.0146	200708	Sa m e

Table C

0. 01 35	0. 00 4	0. 00 2	0. 00 25	0. 00 65	0. 01 02	0. 00 5	0. 01 18	0. 00 8	0	0. 00 45	0. 00 8	0. 01	0. 01 21	30 00 49 1	Sa m e
0. 01 55	0. 00 6	0	0. 00 45	0. 00 85	0. 01 2	0. 00 3	0. 01 38	0. 01	0. 00 2	0. 00 25	0. 00 6	0. 00 8	0. 01 01	20 07 06	Sa m e
0. 01 8	0. 00 85	0. 00 25	0. 00 7	0. 01 1	0. 01 47	0. 00 05	0. 01 63	0. 01 25	0. 00 45	0	0. 00 35	0. 00 55	0. 00 76	30 00 49 2	Sa m e
0. 01 85	0. 00 9	0. 00 3	0. 00 75	0. 01 15	0. 01 52	0	0. 01 68	0. 01 3	0. 00 5	0. 00 05	0. 00 3	0. 00 5	0. 00 71	20 07 14	Sa m e
0. 02 15	0. 01 2	0. 00 6	0. 00 10 5	0. 01 45	0. 01 82	0. 00 3	0. 01 98	0. 01 6	0. 00 8	0. 00 35	0	0. 00 2	0. 00 41	30 00 49 3	Sa m e
0. 02 35	0. 01 4	0. 00 8	0. 01 25	0. 01 65	0. 02 02	0. 00 5	0. 02 18	0. 01 8	0. 01	0. 00 55	0. 00 2	0	0. 00 21	30 00 49 4	Sa m e
0. 02 56	0. 01 61	0. 01	0. 01 46	0. 01 86	0. 02 23	0. 00 71	0. 02 39	0. 00 2	0. 01 21	0. 00 76	0. 00 41	0. 00 21	0	30 00 49 5	Sa m e

Answer each of the following questions in the spaces provided. The answers to the questions and the use of Tables A, B, and C will determine the timing key required to correct the injection timing.

A working example is attached for your review to illustrate the use of this worksheet.

1.	What is the current timing?	0.267_____
2.	What is the timing code?	JF_____
3.	What is the timing specification for this code ($\pm$ 0.002 in)?	0.279_____

4.	Is the current timing a larger or smaller number than the specification? • If larger, advance the timing. • If smaller, retard the timing.	Smaller_____ Retard_____
5.	What is the difference between the current timing (answer to question 1) and the specification (answer to question 3)?	0.012_____
6.	Does the offset of the current key point in the same or opposite direction that the camshaft normally rotates?	Same_____
7.	Use Table A to determine the current key part number. • What is the amount of the offset of the current key? • What is the part number of the current key?	0.007_____ 200711_____

Table A

Timing Key Part Number	Offset	
	mm	in
200704	0.500	0.0197
200706	0.840	0.0331
200708	0.530	0.0210
20709	0.381	0.0150
200711	0.178	0.0070
200714	0.991	0.0390
216294	0.100	0.0040
216782	0.279	0.0110
3000491	0.686	0.0270
3000492	0.914	0.0360
3000493	1.092	0.0430
3000494	1.194	0.0470
3000495	1.295	0.0510

Table A		
Timing Key Part Number	Offset	
	mm	in
3626781	0.229	0.0090
3626782	0.305	0.0120
3626783	0.432	0.0170
3626784	0.546	0.0215
8.	Use Table B to determine how to use Table C. Circle or check the appropriate answer.	

Table B			
Answer to Question 4	Answer to Question 6	Beginning Point on Table C	
Advance	Same	Top of column	_____
Advance	Opposite	Bottom of column	_____
Retard	Same	Bottom of column	_____X_____
Retard	Opposite	Top of column	_____
9.	Answer the following questions BEFORE using Table C to determine the new timing key part number. - Find the current key part number listed at the top of the column on Table C. - Move up or down the column (the answer to question 8). Do not pass 0.000 (zero).		

If you pass 0.000 (zero), you will be choosing a key that does the opposite of what you want it to do.

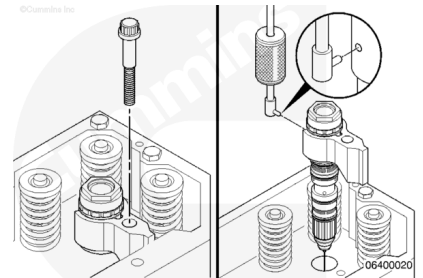
Stop when you locate the number nearest (± 0.002 in) to the required change in push rod travel (answer to question 5). Remain in this row. Move your finger to the right. The result is the new key part number and direction of offset the timing key **must** point.

- What is the part number of the new key? **200706**

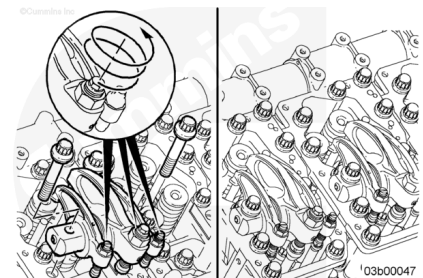
Note : Each column on Table C indicates the change in the push rod travel. The change will result if the key at the top of the column is removed and the new key indicated in the second column from the right is installed. The direction of offset will be pointing as indicated in the last column to the right on Table C.

Finishing Steps

Install the injector from cylinder number 1 on the right bank.
Refer to Procedure 006-026 in Section 6
(</qs3/pubsys2/xml/en/procedures/50/50-006-026-tr.html>).



Install the rocker lever assembly from cylinder number 1 on the right bank. Refer to Procedure 003-009 in Section 3
(</qs3/pubsys2/xml/en/procedures/50/50-003-009.html>).



Install the rocker lever cover from cylinder number 1 on the right bank. Refer to Procedure 003-011 in Section 3
(</qs3/pubsys2/xml/en/procedures/50/50-003-011.html>).

